PH102: Physics-2

Electrodynamics

Lecture # 04

Spherical & Cylindrical Polar Coordinates and Dirac Delta Function

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Previous lecture

We obtained the gradient, divergence and curl in terms of generalized curvilinear coordinates (u_1, u_2, u_3),

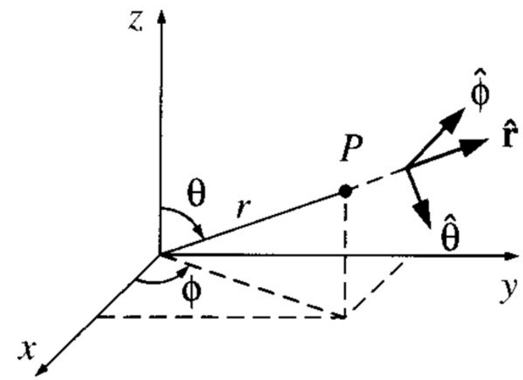
$$\boldsymbol{\nabla} T = \hat{u}_1 \frac{1}{h_1}\frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{h_2}\frac{\partial T}{\partial u_2} + \hat{u}_3 \frac{1}{h_3}\frac{\partial T}{\partial u_3}$$

$$\boldsymbol{\nabla} \cdot \mathbf{v} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}(v_1 h_2 h_3) + \frac{\partial}{\partial u_2}(v_2 h_3 h_1) + \frac{\partial}{\partial u_3}(v_3 h_1 h_2)\right]$$

$$\boldsymbol{\nabla} \times \mathbf{v} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} \hat{u}_1 h_1 & \hat{u}_2 h_2 & \hat{u}_3 h_3 \\ \dfrac{\partial}{\partial u_1} & \dfrac{\partial}{\partial u_2} & \dfrac{\partial}{\partial u_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix}$$

Spherical Polar Coordinates

In spherical polar coordinate system, the three orthogonal coordinates are (r, θ, ϕ) at a given point P as shown in the figure.



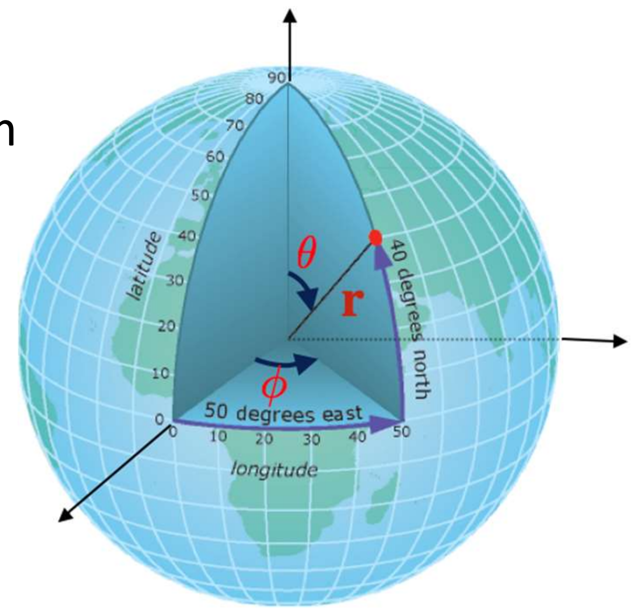
$r \rightarrow$ the radial distance from the origin, i.e. magnitude of position vector.

$\theta \rightarrow$ the angle down the z-axis, i.e. **polar angle**.

$\phi \rightarrow$ the angle around the x-axis, i.e. **azimuthal angle**.

Their relation to Cartesian co-ordinate can be read from the figure

$$x = r\sin\theta\cos\varphi, \quad y = r\sin\theta\sin\varphi, \quad z = r\cos\theta$$



In this coordinate system

$r \rightarrow$ ranges from 0 to ∞, $\theta \rightarrow$ ranges from 0 to π,

$\phi \rightarrow$ ranges from 0 to 2π

(Contd…)

Any vector **A** can be written as the sum of unit vectors, $\mathbf{A} = A_r\hat{\mathbf{r}} + A_\theta\hat{\theta} + A_\phi\hat{\phi}$

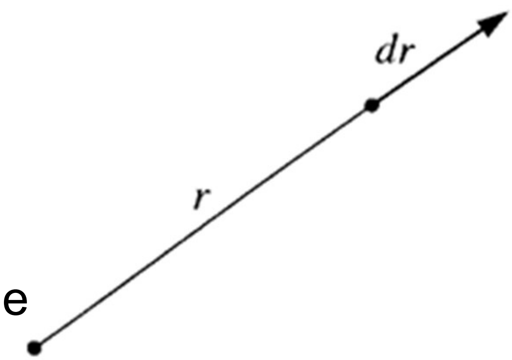
here A_r , A_θ , A_ϕ represent radial, polar and azimuthal components of vector **A.**

The unit vectors in SPC can be written in terms of Cartesian coordinates,

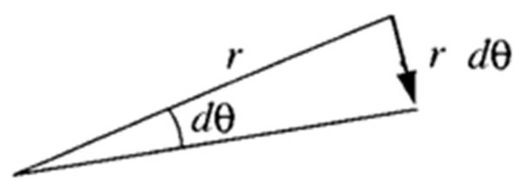
$$\hat{r} = sin\theta\cos\varphi\hat{x} + sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \cos\theta\cos\varphi\hat{x} + \cos\theta\sin\varphi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}$$



Three infinitesimal displacements along directions of three orthogonal coordinates $(\hat{r}, \hat{\theta}, \hat{\phi})$ are shown in the figure.



Here, displacement along radial direction $\hat{r}$ is

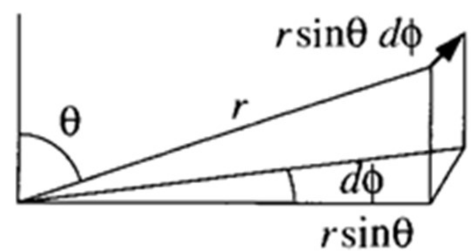
$dl_1 = h_1 du_1 \rightarrow dl_1 = dl_r = dr,$ where $h_1 = 1$

Similarly, along $\hat{\theta}$ direction,

$dl_2 = h_2 du_2 \rightarrow dl_2 = dl_\theta = rd\theta,$ where $h_2 = r$

and along $\hat{\phi}$ direction,

$dl_3 = h_3 du_3 \rightarrow dl_3 = dl_\phi = r\sin\theta\, d\phi,$ where $h_3 = r\sin\theta$



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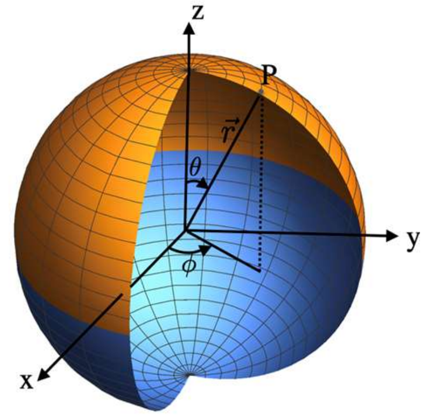
The general infinitesimal displacement is ,

$$d\mathbf{l} = \hat{u}_1 dl_1 + \hat{u}_2 dl_2 + \hat{u}_3 dl_3$$

In spherical polar coordinates, it is written as,

$$d\mathbf{l} = \hat{\mathbf{r}} dl_r + \hat{\theta} dl_\theta + \hat{\phi} dl_\phi = dr\hat{r} + rd\theta\hat{\theta} + r\sin\theta d\phi\hat{\phi}$$

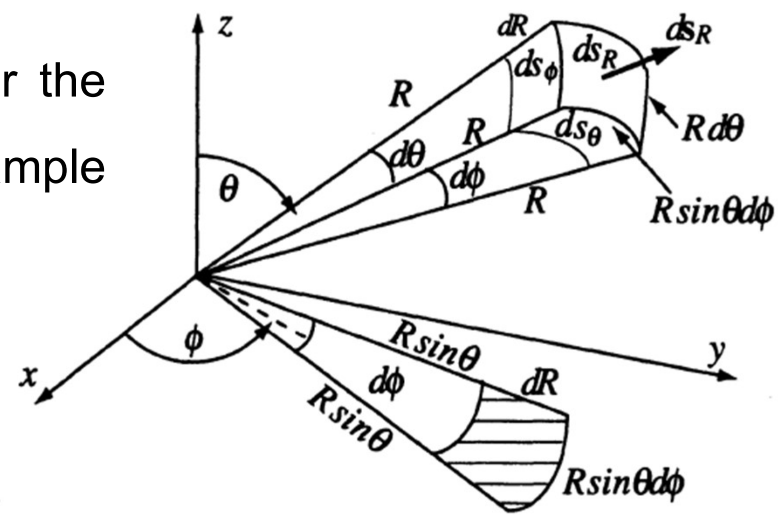


The volume element dτ is the product of three infinitesimal displacements,

$$d\tau = dl_r dl_\theta dl\phi = r^2 \sin\theta\, dr\, d\theta\, d\phi$$

We cannot write down a single expression for the area vector, it depends on the surface. For example if, *r* is fixed, the surface element is,

$$d\mathbf{a}_r = dl_\theta dl_\phi \hat{r} = (r^2 \sin\theta) d\theta\, d\phi \hat{r}$$



Similarly, for other surfaces,

$$d\mathbf{a}_\theta = dl_r dl_\phi \hat{\theta} = (r\sin\theta)\, dr d\phi \hat{\theta}$$

$$d\mathbf{a}_\phi = dl_r dl_\theta \hat{r} = r dr d\theta \hat{r}$$

Gradient in SPC

The next step is to determine the derivatives such as gradient, divergence and curl in SPC system. They can be written straight away by using the expressions derived in OCC and writing down the appropriate h_1 h_2 and h_3 values.

The gradient of a scalar *T*,

$$\nabla T = \sum_i \hat{u}_i \frac{1}{h_i}\frac{\partial T}{\partial u_i} = \hat{u}_1 \frac{1}{h_1}\frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{h_2}\frac{\partial T}{\partial u_2} + \hat{u}_3 \frac{1}{h_3}\frac{\partial T}{\partial u_3}$$

In SPC, $h_1 = 1$, $h_2 = r$, $h_3 = r\sin\theta$, and ($u_1 = r$, $u_2 = \theta$, $u_3 = \phi$) so the gradient is written as,

$$\boxed{\nabla T = \hat{\mathbf{r}}\frac{\partial T}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial T}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial T}{\partial \phi}}$$

Divergence and Curl in SPC

The divergence in OCC system is given by,

$$\nabla \cdot \mathbf{v} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}(v_1 h_2 h_3) + \frac{\partial}{\partial u_2}(v_2 h_3 h_1) + \frac{\partial}{\partial u_3}(v_3 h_1 h_2)\right]$$

and so,

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2 \sin\theta}\left[\frac{\partial}{\partial r}(v_r r^2 \sin\theta) + \frac{\partial}{\partial \theta}(v_\theta r \quad \sin\theta) + \frac{\partial}{\partial \phi}(v_\phi r)\right]$$

or,

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta v_\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(v_\phi)$$

The curl of a vector is,

$$\nabla \times \mathbf{v} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} \hat{u}_1 h_1 & \hat{u}_2 h_2 & \hat{u}_3 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ v_r & r v_\theta & r\sin\theta v_\phi \end{vmatrix}$$

Laplacian in SPC

The Laplacian in SPC can be written by using the expressions for gradient and divergence,

$$\nabla^2 T = \boldsymbol{\nabla} \cdot \boldsymbol{\nabla} T = \boldsymbol{\nabla} \cdot \left[\hat{r}\frac{\partial T}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial T}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial T}{\partial \phi}\right]$$

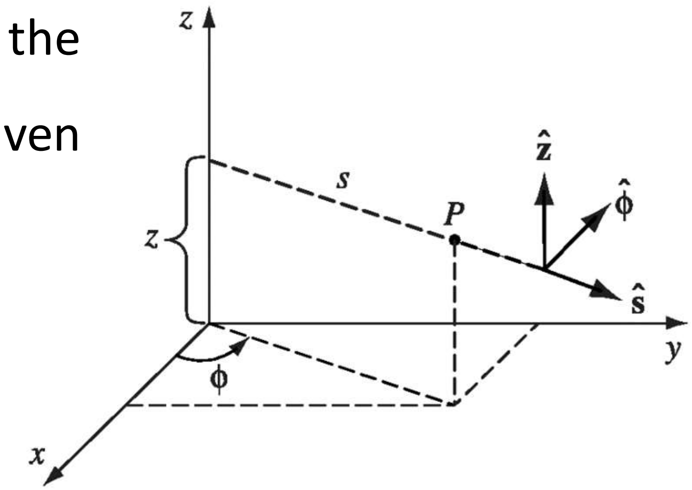
$$\boldsymbol{\nabla} \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta v_\theta) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(v_\phi)$$

$$\nabla^2 T = \boldsymbol{\nabla} \cdot \boldsymbol{\nabla} T \quad = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial T}{\partial r}) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\frac{1}{r}\frac{\partial T}{\partial \theta}) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}(\frac{1}{r\sin\theta}\frac{\partial T}{\partial \phi})$$

$$\nabla^2 T = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial T}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\frac{\partial T}{\partial \theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 T}{\partial \phi^2}$$

Cylindrical Polar Coordinates

In cylindrical polar coordinate (CPC) system, the three orthogonal co-ordinates are (s, ϕ, z) at a given point P as shown in the figure.



$s \rightarrow$ the distance from the z-axis to the point P.

$\phi \rightarrow$ the angle around the x-axis, i.e. **azimuthal angle** (same meaning as in SPC)

$z \rightarrow$ same as that of Cartesian coordinate

Their relation to Cartesian co-ordinate can be read from the figure

$$x = s\cos\varphi, \quad y = s\sin\varphi, \quad z = z$$

In this coordinate system

$s \rightarrow$ ranges from 0 to ∞, $\phi \rightarrow$ ranges from 0 to 2π, $z \rightarrow$ ranges from $-\infty$ 0 to ∞

(Contd…)

Any vector **A** can be written as the sum of unit vectors, $\mathbf{A} = A_s\hat{s} + A_\phi\hat{\phi} + A_z\hat{z}$

here A_s, A_ϕ, A_z represent radial, azimuthal and z components of vector **A.**

The unit vectors can be written in terms of Cartesian coordinates,

$$\hat{s} = \cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}, \qquad \hat{\phi} = -\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}, \qquad \hat{z} = \hat{z}$$

The infinitesimal displacements along three directions are

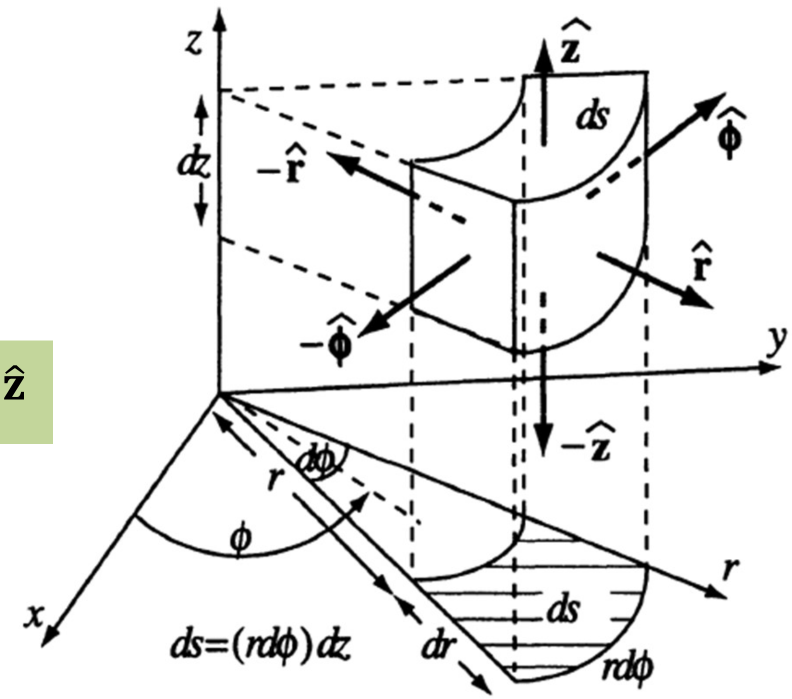
$$dl_1 = h_1 du_1 \rightarrow dl_1 = dl_s = ds$$
$$dl_2 = h_2 du_2 \rightarrow dl_2 = dl_\phi = s d\phi$$
$$dl_3 = h_3 du_3 \rightarrow dl_z = dz$$

where, $h_1 = 1$, $h_2 = s$, $h_3 = 1$

$$d\mathbf{l} = \hat{\mathbf{s}} dl_s + \hat{\phi} dl_\phi + \hat{\mathbf{z}} dl_z = ds\,\hat{\mathbf{s}} + s d\phi\,\hat{\phi} + dz\,\hat{\mathbf{z}}$$

The volume element $d\tau$ is the product of three infinitesimal displacements,

$$d\tau = dl_s dl_\phi dl_z = s ds d\phi dz$$



Read r as s.

Gradient in CPC

The next step is to determine the derivatives such as gradient, divergence and curl in CPC system. They can written straight away by using the expressions derived in OCC and writing down the appropriate h_1, h_2 and h_3 values.

In CPC, $h_1 = 1$, $h_2 = s$, $h_3 = 1$, so the gradient is written as,

$$\boldsymbol{\nabla} T = \sum_i \hat{u}_i \frac{1}{h_i}\frac{\partial T}{\partial u_i} = \hat{u}_1 \frac{1}{h_1}\frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{h_2}\frac{\partial T}{\partial u_2} + \hat{u}_3 \frac{1}{h_3}\frac{\partial T}{\partial u_3}$$

$$\boxed{\boldsymbol{\nabla} T = \hat{\mathbf{s}}\frac{\partial T}{\partial s} + \hat{\phi}\frac{1}{s}\frac{\partial T}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}}$$

Divergence and Curl in CPC

The divergence in OCC is given by,

$$\nabla \cdot \mathbf{v} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}(v_1 h_2 h_3) + \frac{\partial}{\partial u_2}(v_2 h_3 h_1) + \frac{\partial}{\partial u_3}(v_3 h_1 h_2)\right]$$

and correspondingly,

$$\nabla \cdot \mathbf{v} = \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_s) + \frac{\partial}{\partial \phi}(v_\phi) + \frac{\partial}{\partial z}(s v_z)\right] = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

The curl is

$$\nabla \times \mathbf{v} = \hat{\mathbf{s}}\left[\frac{1}{s}\frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}\right] - \hat{\phi}\left[\frac{\partial v_z}{\partial s} - \frac{\partial v_s}{\partial z}\right] + \hat{\mathbf{z}}\frac{1}{s}\left[\frac{\partial v_s}{\partial \phi} - \frac{\partial v_\phi}{\partial s}\right]$$

$$\nabla \times \mathbf{v} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} \hat{u}_1 h_1 & \hat{u}_2 h_2 & \hat{u}_3 h_3 \\ \dfrac{\partial}{\partial u_1} & \dfrac{\partial}{\partial u_2} & \dfrac{\partial}{\partial u_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix} = \frac{1}{s}\begin{vmatrix} \hat{\mathbf{s}} & s\hat{\phi} & \hat{\mathbf{z}} \\ \dfrac{\partial}{\partial s} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ v_s & s v_\phi & v_z \end{vmatrix}$$

Laplacian in CPC

The Laplacian in CPC can be written by using the expressions for gradient and divergence,

$$\nabla T = \hat{\mathbf{s}}\frac{\partial T}{\partial s} + \hat{\phi}\frac{1}{s}\frac{\partial T}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}$$

$$\nabla \cdot \mathbf{v} = \frac{1}{s}\left[\frac{\partial}{\partial s}(s v_s) + \frac{\partial}{\partial \phi}(v_\phi) + \frac{\partial}{\partial z}(s v_z)\right] = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$$

$$\nabla^2 T = \nabla \cdot \nabla T = \frac{1}{s}\frac{\partial}{\partial s}(s v_s) + \frac{1}{s}\frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z} = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial T}{\partial s}\right) + \frac{1}{s}\frac{\partial}{\partial \phi}\left(\frac{1}{s}\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(\frac{\partial T}{\partial z}\right)$$

or,

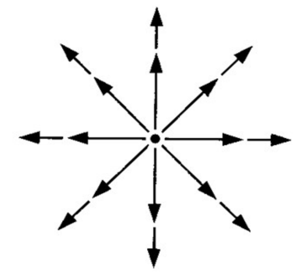
$$\nabla^2 T = \frac{1}{s}\frac{\partial}{\partial s}\left(s\frac{\partial T}{\partial s}\right) + \frac{1}{s^2}\frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2}$$

Is the divergence theorem false???

Let us verify the divergence theorem for the vector field $\vec{V} = \frac{1}{r^2}\hat{r}$

The divergence of $\vec{V}$ is $\vec{\nabla}\cdot\vec{V} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}(v_1 h_2 h_3) + \frac{\partial}{\partial u_2}(v_2 h_3 h_1) + \frac{\partial}{\partial u_3}(v_3 h_1 h_2)\right]$

$v_1 = \frac{1}{r^2}, v_2 = 0, v_3 = 0, h_1 = 1, h_2 = r, h_3 = r\sin\theta\,.$



$$\vec{\nabla}\cdot\vec{V} = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta.\frac{1}{r^2}\right)\right] = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{1}{r^2}\right) = \frac{1}{r^2}\frac{\partial}{\partial r}(1) = 0$$

$$\boxed{\int_v \vec{\nabla}\cdot\vec{V}d\tau = \oint_s \vec{V}\cdot d\vec{a}}$$

Therefore, $\int_v \vec{\nabla}\cdot\vec{V}\,d\tau = 0$

Suppose we integrate over a sphere of radius *R*, centered at the origin; the surface integral is

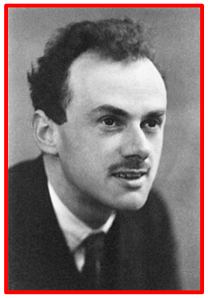
$$\oint_s \vec{V}\cdot d\vec{a} = \int_S \left(\frac{1}{R^2}\hat{r}\right)\cdot\left(R^2\sin\theta\,d\theta d\varphi\hat{r}\right) = \left(\int_0^{\pi}\sin\theta\,d\theta\right)\left(\int_0^{2\pi}d\varphi\right) = 4\pi, \quad d\vec{a} = R^2\sin\theta\,d\theta d\varphi\hat{r}$$

$$\int_v \vec{\nabla}\cdot\vec{V}d\tau \neq \oint_s \vec{V}\cdot d\vec{a}$$

We expected the two results to be the same !!!

(Contd...)

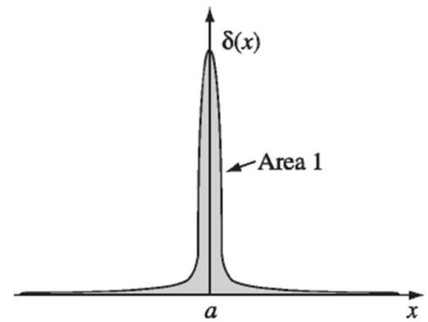
- $\begin{cases} \vec{\nabla} \cdot \vec{V} = 0, & r \neq 0 \\ \vec{\nabla} \cdot \vec{V} = indeterminate, & r = 0 \end{cases}$
- The surface integral is independent of *R*; if the divergence theorem is right (and it is), we should get $\int_v \vec{\nabla} \cdot \vec{V}\, d\tau = 4\pi$ for any sphere centered at the origin, no matter how small.
- Evidently, the entire contribution must be coming from the point $r = 0$!
- Thus, $\vec{\nabla} \cdot \vec{V}$ has the bizarre property that it vanishes everywhere except at one point, and yet its integral (over any volume containing that point) is 4π. No ordinary function behaves like that (density, defined as mass per unit volume, is another quantity that behaves identically).
- **What we have stumbled on is a mathematical object known to physicists as the Dirac delta function.**



Paul Adrien Maurice Dirac
1933 Nobel Prize "for the discovery of new productive forms of atomic theory"

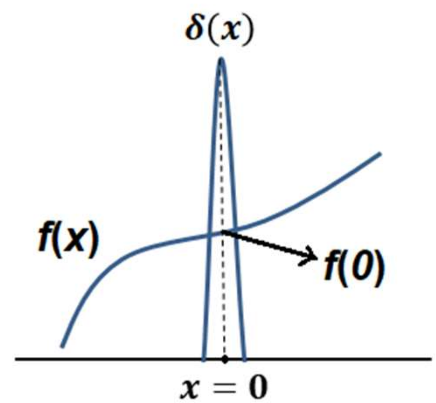
The Dirac delta function

The one dimensional Dirac delta function, $\delta(x)$, can be pictured as an infinitely high, infinitesimally narrow "spike," with area 1.



$$\delta(x) = \begin{Bmatrix} 0, \, if \; x \neq 0 \\ \infty, if \; x = 0 \end{Bmatrix}, \quad \int_{-\infty}^{\infty} \delta(x)\, dx = 1$$

If $f(x)$ is some "ordinary" and continuous function, then the product $f(x)\delta(x)$ = 0 everywhere except at x = 0. It follows that $f(x)\ \delta(x) = f(0)\ \delta(x)$. Since the product is zero anyway except at x = 0, we may as well replace $f(x)$ by the value it assumes at the origin. Therefore, we can write

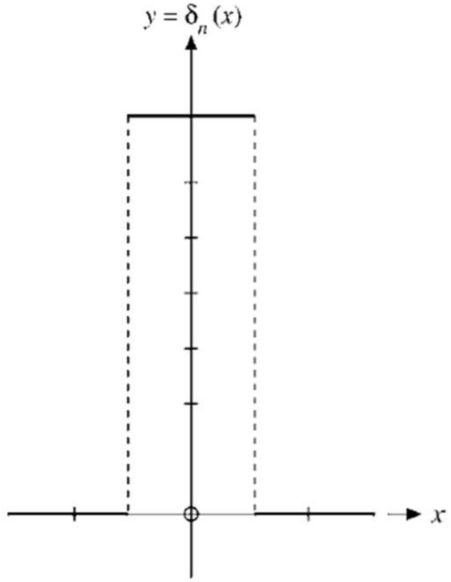


$$\int_{-\infty}^{\infty} f(x)\delta(x)\, dx = f(0) \int_{-\infty}^{\infty} \delta(x)\, dx = f(0).$$

Under an integral, then, the delta function "picks out" value of $f(x)$ at x = 0. (Here the integral need not run from - ∞ to + ∞; it is sufficient that the domain extend across the delta function.

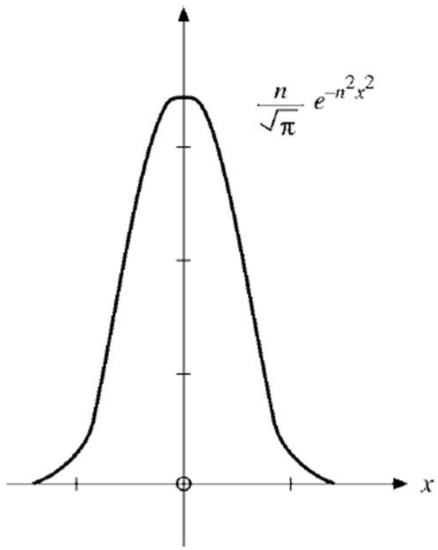
(Contd…)

The term "***Delta function***" is a misnomer. Its value is not finite at x = 0. One can, however, think of delta function as the limit of a sequence of a function. In the mathematical literature it is known as a **generalized function**, or a **distribution**.



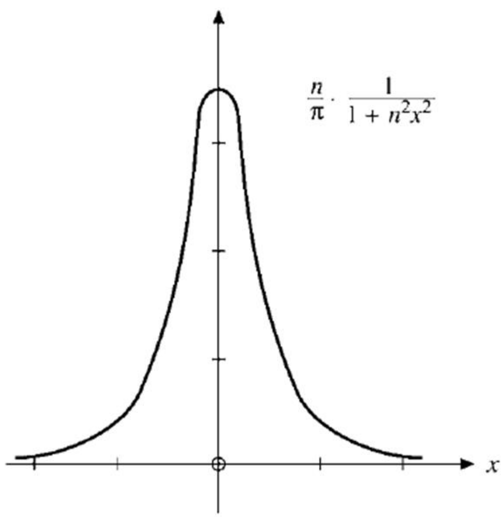
$$\delta_n(x) = \begin{cases} 0, & x < -\frac{1}{2n} \\ n, & -\frac{1}{2n} < x < \frac{1}{2n} \\ 0, & x > \frac{1}{2n} \end{cases}$$

Rectangular distribution function



$$\delta_n(x) = \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2)$$

Gaussian distribution function



$$\delta_n(x) = \frac{n}{\pi} \cdot \frac{1}{1 + n^2 x^2}$$

Lorentzian distribution function

(Contd…)

All the sequences are symmetric (even functions of x).

These are **sequences** of normalized functions: $\int_{-\infty}^{\infty} \delta_n(x)\, dx = 1.$

The sequence of integrals has the limit, $\lim_{n\to\infty} \int_{-\infty}^{\infty} \delta_n(x) f(x)\, dx = f(0).$

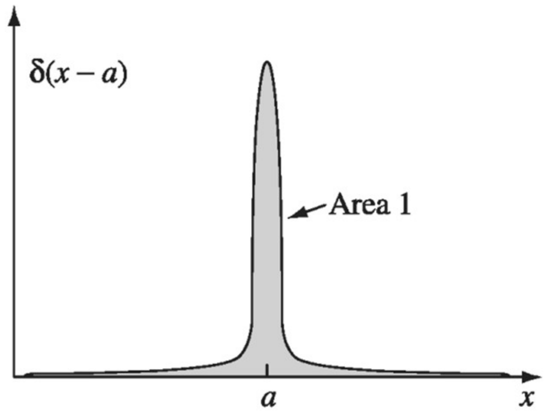
As the limit is approached , the height of the peak increases and the width shrinks in such a way that the area under the curve remains unity.

Note that the limit of $\delta_n(x)$, $n \to \infty$, does not exist. (The limits for all three forms of $\delta_n(x)$ diverge at x = 0.).

We may treat $\delta(x)$ in the form (if δ is represented as a distribution)

$$\int_{-\infty}^{\infty} \delta(x) f(x)\, dx = \lim_{n\to\infty} \int_{-\infty}^{\infty} \delta_n(x) f(x)\, dx.$$

We can shift the spike from *x* = 0 to some other point *x* = *a* and we can write



$$\delta(x-a) = \begin{Bmatrix} 0, \, if \ x \neq a \\ \infty, if \ x = a \end{Bmatrix}, \qquad \int_{-\infty}^{\infty} \delta(x-a)\, dx = 1$$

Accordingly,

$$f(x)\delta(x-a) = f(a)\delta(x-a), \qquad \int_{-\infty}^{\infty} f(x)\delta(x-a)\, dx = f(a)$$

It's best to think of the delta function as something that is *always intended for use under an integral sign.*

The delta function is frequently invoked to describe very short-range forces, such as nuclear forces. It also appears in the normalization of continuum wave functions of quantum mechanics.

Three dimensional Dirac delta function

The three dimensional delta function is $\delta^3(\mathbf{r}) = \delta(x)\delta(y)\delta(z)$

$$\delta^3(\mathbf{r}) = \delta^3(x,y,z) = \begin{Bmatrix} 0 & if\ x^2+y^2+z^2 \neq 0 \\ \infty & if\ x^2+y^2+z^2 = 0 \end{Bmatrix}$$

This three-dimensional delta function is zero everywhere except at (0, 0, 0), where it blows up. Its volume integral is 1:

$$\int_{All\ space} \delta^3(\mathbf{r})d\tau = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \delta(x)\delta(y)\delta(z)\,dxdydz = 1$$

$\delta^3(\mathbf{r} - \mathbf{r}_o)$ can be written as the product of three one dimensional functions

$$\delta^3(\mathbf{r} - \mathbf{r_o}) = \delta(x - x_o)\delta(y - y_o)\delta(z - z_o)$$

$$\int f(\mathbf{r})\delta^3(\mathbf{r} - \mathbf{r_o}) = f(\mathbf{r_o})$$

When we shift the origin from $x = 0$ to $x = a$

$$\int_{All\ space} f(\mathbf{r})\delta^3(\mathbf{r} - \mathbf{a})d\tau = f(\mathbf{a})$$

Resolving the divergence theorem paradox

Based on the discussion till now, we can write,

$$\nabla \cdot (\frac{\hat{r}}{r^2}) = \begin{cases} 0, r \neq 0 \\ 4\pi, r = 0 \end{cases}$$

or, more precisely, using the definition of Dirac delta function, we can write

$$\nabla \cdot (\frac{\hat{r}}{r^2}) = 4\pi\delta^3(\mathbf{r})$$ where, $\delta^3(\mathbf{r}) = \delta(x)\delta(y)\delta(z)$

Therefore,

$$\int \nabla \cdot (\frac{\hat{r}}{r^2}) d\tau = \int_{-\infty}^{+\infty} 4\pi\delta^3(\mathbf{r}) d\tau = 4\pi$$

With the notation $\boldsymbol{\mathscr{r}} \equiv \mathbf{r} - \mathbf{r}'$, we can show that

$$\nabla \left(\frac{1}{\mathscr{r}}\right) = -\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}$$

$$\nabla^2 \frac{1}{\mathscr{r}} = -4\pi\delta^3(\boldsymbol{\mathscr{r}}).$$

Properties of δ functions

i. $\delta(-x) = \delta(x)$

ii. $x\delta(x) = 0$

iii. $\delta(ax) = \frac{1}{|a|}\delta(x), if a \neq 0$

iv. $f(x)\delta(x-a) = f(a)\delta(x-a)$

v. $\int_{-\infty}^{\infty} \delta'(x) f(x)\, dx = -f'(0)$

vi. $\delta'(-x) = -\delta'(x)$

Solved example 1

Find the volume of a sphere of radius R.

Solution:

$$\begin{aligned} V &= \int d\tau = \int_{r=0}^{R} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^2 \sin\theta \, dr \, d\theta \, d\phi \\ &= \left(\int_0^R r^2 \, dr\right) \left(\int_0^{\pi} \sin\theta \, d\theta\right) \left(\int_0^{2\pi} d\phi\right) \\ &= \left(\frac{R^3}{3}\right)(2)(2\pi) = \tfrac{4}{3}\pi R^3. \end{aligned}$$

Solved example 2

Evaluate the integral

$$\int_0^3 x^3 \delta(x-2)\, dx.$$

We know that

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)\, dx = f(a).$$

Comparing $\int_{-\infty}^{\infty} f(x)\delta(x-a)\, dx$ and $\int_0^3 x^3\delta(x-2)\, dx$ we have $f(x) = x^3,\; a = 2$

$$\int_0^3 x^3\delta(x-2)\, dx = \int_{-\infty}^{0} x^3\delta(x-2)\, dx + \int_0^3 x^3\delta(x-2)\, dx + \int_3^{\infty} x^3\delta(x-2)\, dx$$

$$= 0 + 2^3 + 0$$

$$= 8$$

Solved example 3

Evaluate the integral $J = \int_v (r^2+2)\, \nabla . \left(\frac{\hat{\mathrm{r}}}{r^2}\right) d\tau$ where V is a sphere of radius *R'* centered at the origin.

Using Delta function, $$J = \int_v (r^2+2)\, 4\pi\delta^3(\mathrm{r}) d\tau = 4\pi(0+2) = 8\pi$$

Using the product rule, $$\int_v f(\nabla . \mathbf{A})\, d\tau = -\int_v \mathbf{A} . (\nabla f) d\tau + \oint_S f\, \mathbf{A} . d\mathbf{a}$$

The above integral can be written as $$J = -\int \frac{\hat{r}}{r^2} . [\nabla(r^2+2)] d\tau + \oint (r^2+2) \frac{\hat{r}}{r^2} . d\mathbf{a}$$

The gradient is $$\nabla(r^2+2) = 2r\hat{r}$$

so the volume integral becomes $$\int \frac{2}{r} d\tau = \frac{2}{r} R^2 \sin\theta\, dr d\theta d\varphi = 8\pi \int_0^{R'} R dR = 4\pi\, R'^2$$

On the boundary of the sphere, *r* = *R*', therefore $d\mathbf{a} = R'^2 \sin\theta\, d\theta d\varphi \hat{r}$

The surface integral becomes $$\int \left(R'^2+2\right) \sin\theta\, d\theta d\varphi = 4\pi R'^2$$

Total value of J is $$J = -4\pi R'^2 + 4\pi\left(R'^2+2\right) = 8\pi$$

Questions?